

A Cascade of Affine-Modulated Transformers for Cued Change Detection

Two affine self-attention stages with a split actor/critic readout — a mechanistic deep dive on the `affine_cascade` model

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Abstract

We give a mechanistic deep dive on **`affine_cascade`**, a recurrent vision transformer for cued orientation-change detection built from *two* affine-modulated self-attention stages and *two* spatial LSTM memories, read out by a *split* actor/critic head: the first stage’s memory H_1 feeds the critic and the second stage’s memory H_2 feeds the actor. Trained from sparse reward on the seven-step, four-stimulus task with a frozen pretrained VAE front-end (snapshot at 24,000 iterations), the agent detects changes at near-ceiling for large changes (cued hit 0.985, false-alarm 0.083) and shows a reliability-scaled spatial cueing benefit (cued-versus- uncued threshold advantage growing from $+1.1^\circ$ at 25% to $+4.9^\circ$ at 75% cue reliability). Four mechanistic analyses reveal a clean **stage dissociation**: (i) the first transformer *orients to the cue* — its attention to the cued patch swaps with the cue (cue@ $S_1 \rightarrow S_1$, cue@ $S_4 \rightarrow S_4$) and climbs to ≈ 0.8 during the stimulus even with no change, with a further bump to 0.93 when a change occurs there — whereas the second transformer’s attention is position-biased and does not track the cue (cue-following index $+0.66$ vs $+0.02$); (ii) the *critic’s* memory H_1 maintains the cue location at ceiling (1.00) for the entire trial, whereas the *actor’s* memory H_2 lets it decay to ≈ 0.6 ; (iii) the critic is modestly calibrated and its outcome uncertainty falls with evidence ($0.28 \rightarrow 0.06$); and (iv) — unlike the lab’s split-readout cross-talk model — biasing either transformer’s attention is a *weak* causal lever on behaviour, indicating that `affine_cascade` computes through feature-wise affine modulation (Γ, β) rather than through attention reallocation. The picture is of a two-stage agent whose first stage does the perceptual work and holds the cue, and whose second stage reads a processed decision signal.

1. Introduction

The recurrent vision transformer parses each video frame into spatial tokens, runs a recurrent self-attention block whose attention is computed jointly from the image and a working memory, writes that memory with a spatial LSTM, and reads it with a distributional actor-critic head. The **affine_cascade** variant studied here makes two design choices at once: it stacks *two* such stages in a cascade, and it *splits* the readout so the two streams feed different heads. Each stage’s attention is **affine-modulated** — the memory derives a feature scale-matrix Γ and shift β that transform the image before a standard self-attention ($X' = \Gamma(H)X + \beta(H)$). This report asks what each stage computes, where the cue is held, which stream carries the decision, and whether attention is the causal lever.

1.1 The architecture in one paragraph

Per frame, stage one applies affine self-attention to the VAE tokens X with feedback from its own memory, $Z_1 = \text{SA}_{\text{aff}}(X; H_1)$, and writes $H_1 \leftarrow \text{LSTM}_1(Z_1)$; stage two applies affine self-attention to Z_1 with feedback from H_2 , $Z_2 = \text{SA}_{\text{aff}}(Z_1; H_2)$, and writes $H_2 \leftarrow \text{LSTM}_2(Z_2)$. The actor reads the flattened H_2 ; the critic reads the flattened H_1 . By construction H_1 is independent of stage two, so the critic pathway is a strict prefix of the actor pathway.

2. Methods

2.1 Task and front-end

The seven-step cued change-detection task presents a blank frame, a spatial cue (a ring whose completeness encodes reliability) at one of two locations, a blank, then four noisy Gabor quadrants; on half the trials one quadrant changes orientation at $t = 5$. Reward is uniform (+1 for a correct detection after the change or a correct rejection) and the cue is achromatic, so — unlike the long-task v-series — there is *no* value-by-colour structure to report. The front-end is a reconstruction-pretrained, frozen patch-VAE producing four 140-d tokens.

2.2 Faithful extraction and intervention

All analyses use the model’s own forward step. The two attention maps aw_1, aw_2 (one per transformer, each 4×4) and the two memories H_1, H_2 are read directly from the recurrent state. For the causal experiment we add a logit bias to a chosen patch’s attention column in either transformer ($\text{attn_clamp} = \{\text{t1}|\text{t2} : \{j : b\}\}$) and sweep b .

2.3 Analyses

Behaviour (psychometric/chronometric by cue reliability, cued vs. uncued); attention allocation (per-frame spatial maps and cued-patch timecourses for both transformers); temporal linear decoding of task variables from H_1 vs. H_2 (balanced-accuracy logistic readouts, 3-fold CV); causal attention biasing; and value/critic calibration.

3. Results

3.1 Behaviour: a reliability-scaled cueing benefit

The agent solves the task: the cued psychometric function is steep and saturates near 1.0 for large changes (cued hit 0.985 at $\Delta = 64^\circ$), with a false-alarm rate of 0.083 (Figure 1). The cued curve

sits left of the uncued one — a spatial cueing benefit — and the threshold advantage grows with displayed cue reliability, from $+1.1^\circ$ at 25% to $+2.6^\circ$ at 50% to $+4.9^\circ$ at 75%. At 100% reliability the measured benefit falls back to $+1.1^\circ$; this is an artefact of the protocol rather than the model, because a perfectly reliable cue makes *uncued* change trials off-distribution. Reaction time decreases mildly with change magnitude.

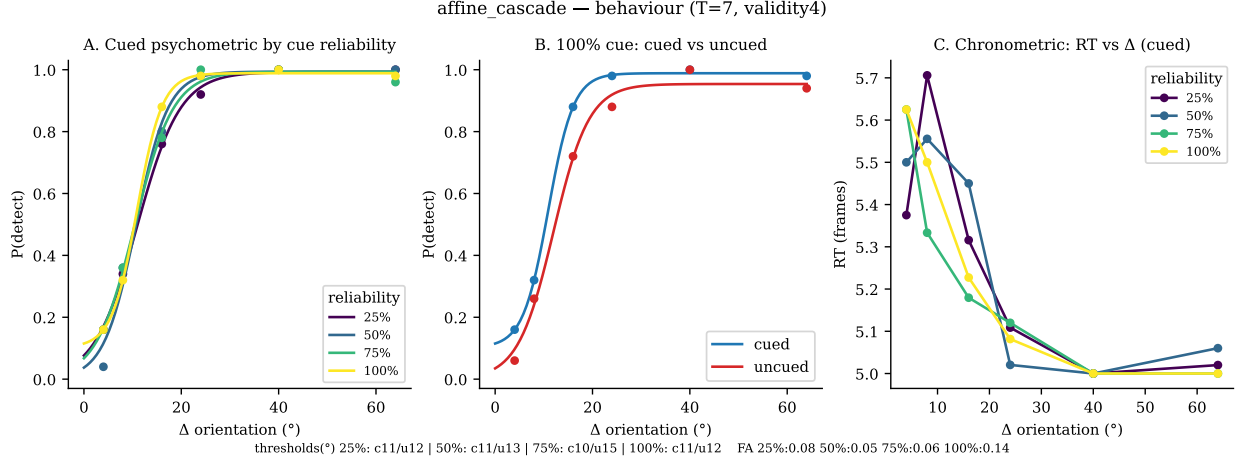


Figure 1: Behaviour. **A:** cued psychometric by displayed cue reliability. **B:** cued vs. uncued at 100% reliability. **C:** chronometric reaction time vs. change magnitude. Thresholds and false-alarm rates are printed beneath; the cued advantage grows from 25% to 75% reliability.

3.2 Attention allocation: stage one orients to the cue, stage two does not

The two transformers allocate attention very differently (Figures 2, 3). **Stage one orients to the cued location.** With the cue forced at S_1 , its attention to S_1 climbs through the stimulus period to 0.79 *even on no-change trials*; move the cue to S_4 and the peak follows (attention to S_4 reaches 0.79, while attention to S_1 falls to 0.13) — a cue-following index of +0.66. An actual change at the attended location adds a further bump (to 0.93). This is a genuine cue-orienting attention, and it is notable: the lab’s multiplicatively-gated model is change-locked and does *not* orient on this task. **Stage two does not track the cue.** Its attention is position-biased: at the cue/blank frames it fixates a single quadrant (S_4 for a left cue; the top-right quadrant for a right cue) regardless of where the cue actually is, and is diffuse thereafter (cue-following index +0.02).

3.3 What the two memories hold: the critic’s memory keeps the cue

Temporal linear decoding (Figure 4) reveals the deepest dissociation. The **cue location** decodes at ceiling from *both* memories at cue onset, but only the **critic’s** memory H_1 *maintains* it — staying at ≈ 1.00 through the entire delay and stimulus period — while the **actor’s** memory H_2 lets it decay to ≈ 0.6 by the decision. Cue reliability is decodable only transiently around the cue. The changed location is decoded only weakly from either memory (≤ 0.45 , chance 0.25). So the cue prior lives in the first-stage memory that feeds the critic, not in the actor’s memory; the actor reads a representation from which the raw cue has largely washed out.

affine_cascade — attention maps, 100% cue@S1 (T1→H1→critic, T2→H2→actor; cyan=cued/changed S1)

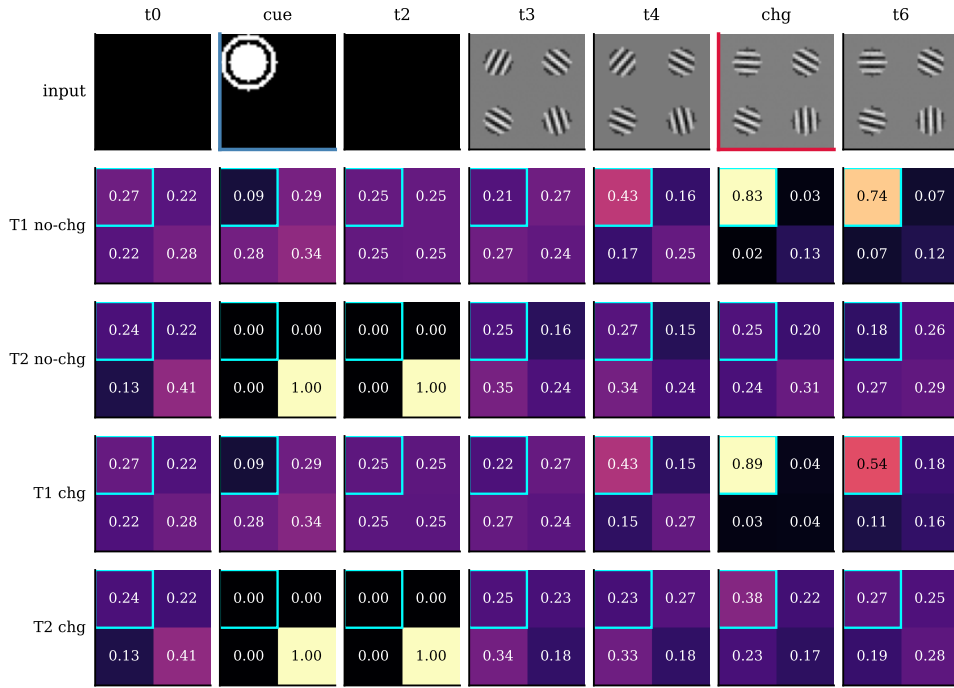


Figure 2: Per-frame attention maps for both transformers, 100% cue at S_1 (top-left, cyan), no-change and change conditions; scores printed in each 2×2 cell. Stage one (T1→ H_1 →critic) concentrates on the cued quadrant during the stimulus (0.79 with no change) and reaches 0.93 when a change occurs there; stage two (T2→ H_2 →actor) fixates an off-cue quadrant at the cue/blank frames (here 1.00 on S_4) and is diffuse later.

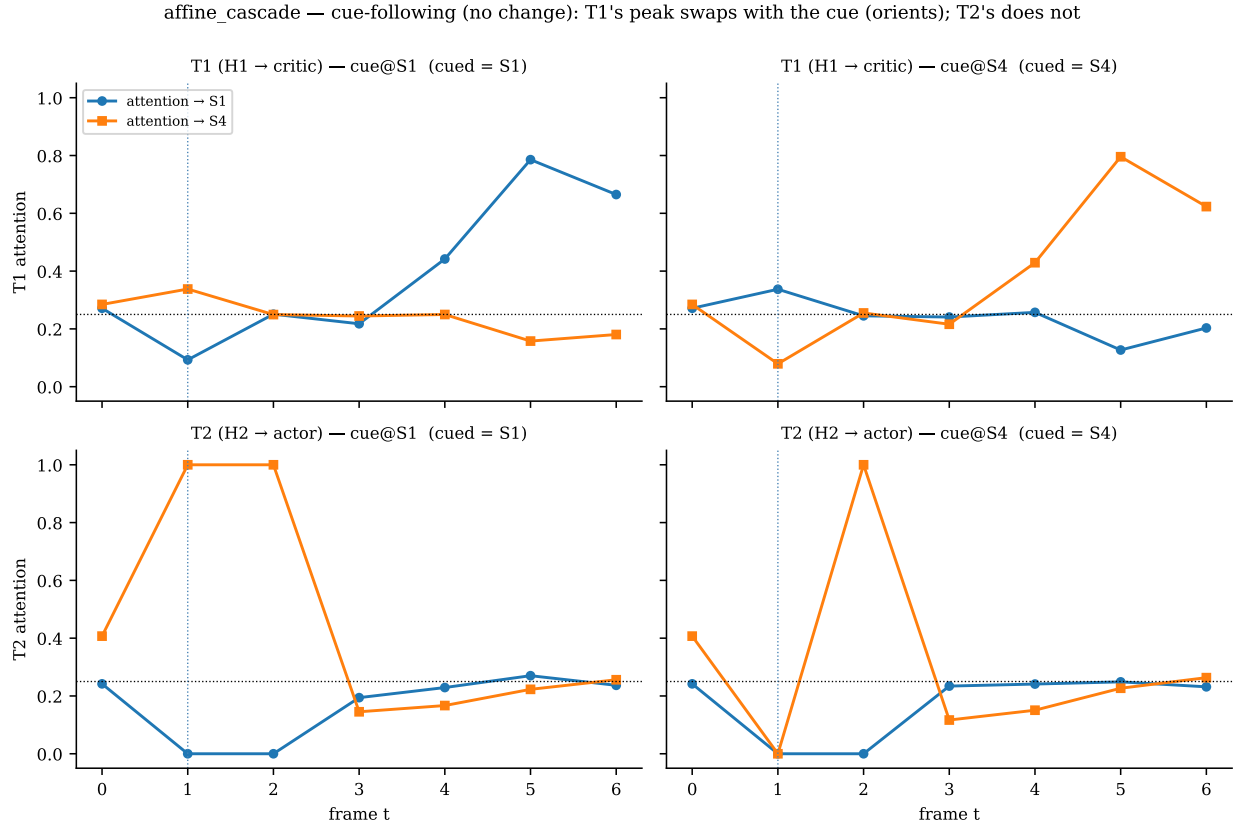


Figure 3: Cue-following (no change): attention to S_1 vs. S_4 over the trial, per transformer (rows) and cue position (columns). Stage one's peak *swaps* with the cue — it orients; stage two's peak stays on the same quadrant regardless of the cue — it does not follow. Same forced-cue protocol as Figure 2.

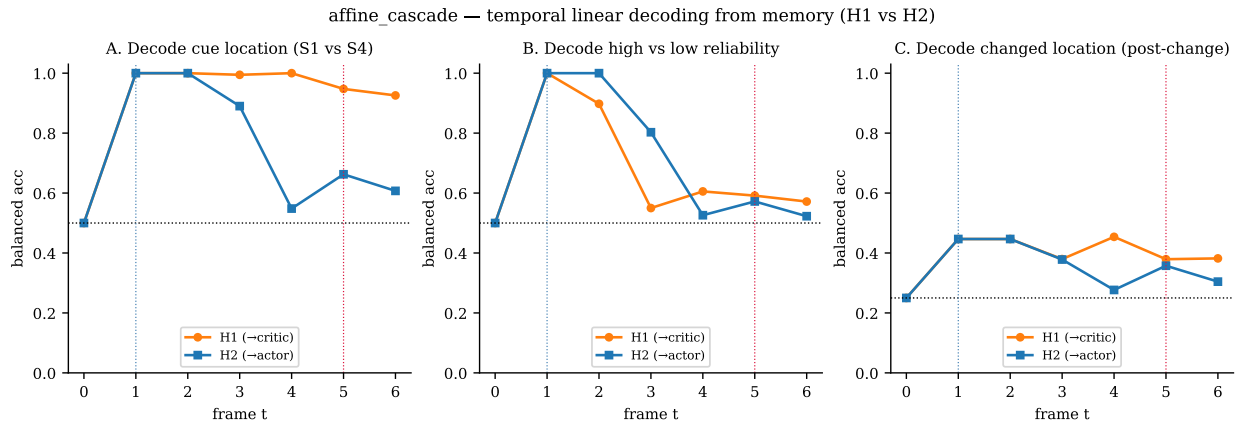


Figure 4: Temporal linear decoding of task variables from memory, H_1 (critic) vs. H_2 (actor). **A:** cue location is *maintained* at ceiling in H_1 but decays in H_2 . **B:** reliability decodes transiently. **C:** the changed location is only weakly linearly available in either memory.

3.4 Causal attention: a weak lever — computation is in the modulation, not the map

Biasing either transformer’s attention toward the changed quadrant over $b \in [-6, 6]$ moves behaviour only slightly (Figure 5). Stage-one bias has a small *negative* effect on the hit rate ($0.917 \rightarrow 0.867$ as bias goes $0 \rightarrow +6$; -0.09 over the full sweep); stage-two bias has no consistent effect (± 0.05 , no trend); and neither bias moves the critic’s outcome uncertainty (flat at ≈ 0.12). This is markedly different from the lab’s split-readout *cross-talk* model, where salience clamps strongly move the decision and top-down clamps move value. The natural reading is mechanistic: *affine_cascade* does its work through the *feature-wise affine modulation* $\Gamma(H)X + \beta(H)$ applied before attention, so reweighting the post-modulation attention map is a weak intervention. The decision is robust to where attention points.

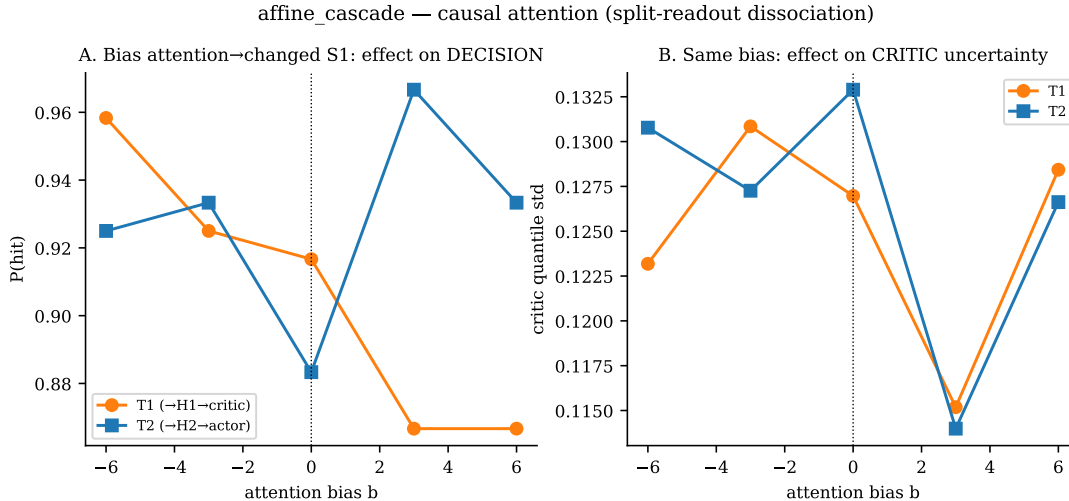


Figure 5: Causal attention. **A:** biasing each transformer’s attention toward the changed quadrant barely moves the decision (hit rate). **B:** and does not move the critic’s uncertainty. Attention reallocation is a weak lever in this model, unlike the cross-talk variant.

3.5 A calibrated critic whose uncertainty tracks evidence

The distributional critic (which reads H_1) is modestly calibrated: per-step predicted value correlates with the realised discounted return ($r = 0.22$), state value rises over the trial as evidence accumulates, and — the cleanest value signature — the critic’s outcome uncertainty (quantile spread just after the change) falls monotonically as the change becomes more detectable (0.28 at $\Delta = 4^\circ \rightarrow 0.06$ at $\Delta = 64^\circ$; Figure 6). More detectable changes leave the value function more certain about the outcome.

4. Discussion

A two-stage division of labour. The results cohere into a single account. The first stage — its attention, its memory H_1 , and the critic it feeds — does the *perceptual* work: it *orients* attention to the cued quadrant (the peak tracks the cue) and holds the cue location at ceiling for the whole trial. The second stage — H_2 and the actor — reads a *processed* signal in which the raw cue has decayed, and issues the decision. That the **critic** ends up holding the stimulus while the **actor** holds a more abstract code is the opposite of the naive expectation, and follows directly from

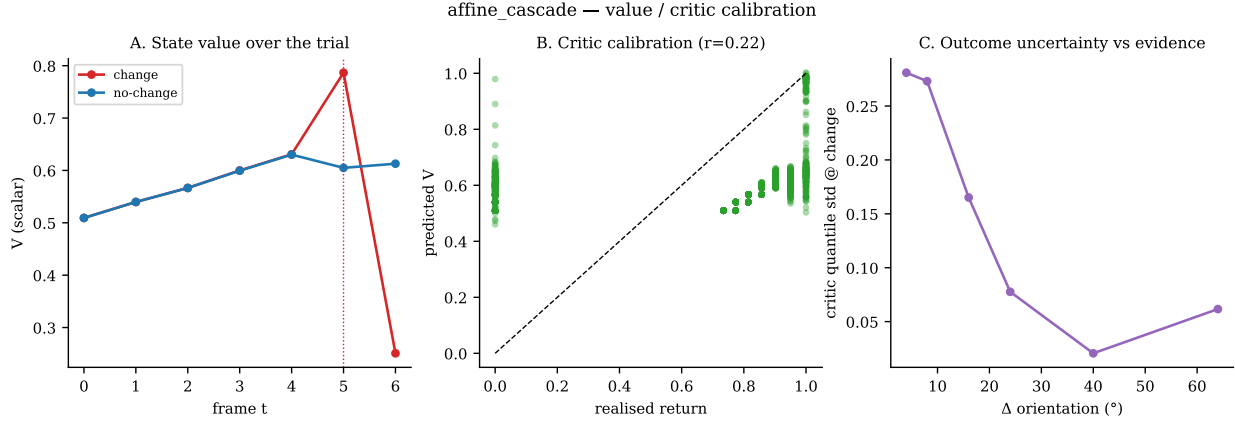


Figure 6: Value / critic. **A:** state value over the trial, change vs. no-change. **B:** critic calibration (per-step predicted value vs. realised return). **C:** outcome uncertainty falls as the change grows more detectable.

the cascade: because the actor’s memory is two transformations downstream of the image, the linearly-recoverable stimulus content has been transformed away by the time it reaches H_2 .

Stage one orients to the cue — but the orienting is a correlate, not the lever. Unlike the multiplicatively-gated model, *affine_cascade* *does* produce a cue-orienting attention map (stage one, §3.2). Yet biasing that map barely moves behaviour (§3.4). These are reconciled by where the computation lives: the cue prior is held in the memory H_1 (decodable at ceiling all trial), the stimulus is transformed by the feature-wise affine modulation Γ, β , and the attention map is a *readout* of that orienting rather than the causal bottleneck. So the cueing benefit is memory-borne and the decision is robust to attention clamps, even though the attention visibly follows the cue — the map reflects the computation without gating it.

Limitations. This is a snapshot at 24,000 of 50,000 iterations (cued hit already 0.985 at large Δ); the analysis should be re-run at convergence. The 100%-reliability cueing point is confounded by off-distribution uncued trials. The causal null is a property of this model’s mechanism, but the clamp acts only on the post-modulation attention; a complementary intervention on Γ, β would localise the computation more directly and is the natural next experiment.

5. Conclusion

affine_cascade learns cued change detection well and exposes a clean two-stage dissociation: a first stage that orients attention to the cue and holds it for the critic, and a second stage that reads a processed code for the actor. Its cueing benefit is memory-borne and reliability-scaled, its critic is calibrated with evidence-tracking uncertainty, and — distinctively — its decision is computed through affine feature modulation rather than attention reallocation, making the attention map a weak causal lever. The model is a strong, interpretable candidate whose mechanism differs informatively from the multiplicative and cross-talk variants.